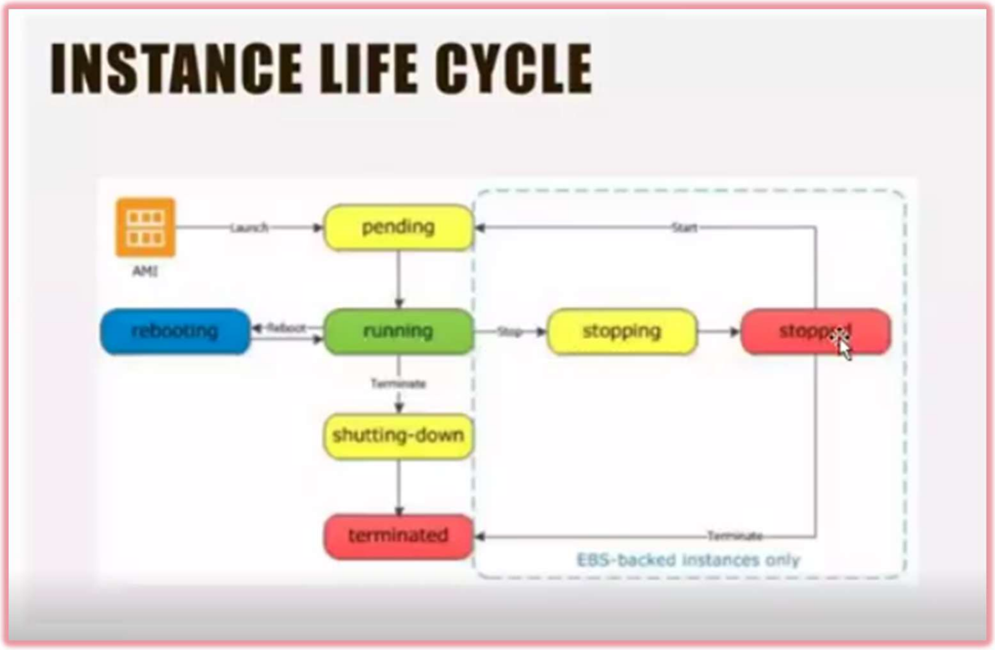
Aws – Jovac 2026

Lecture 1

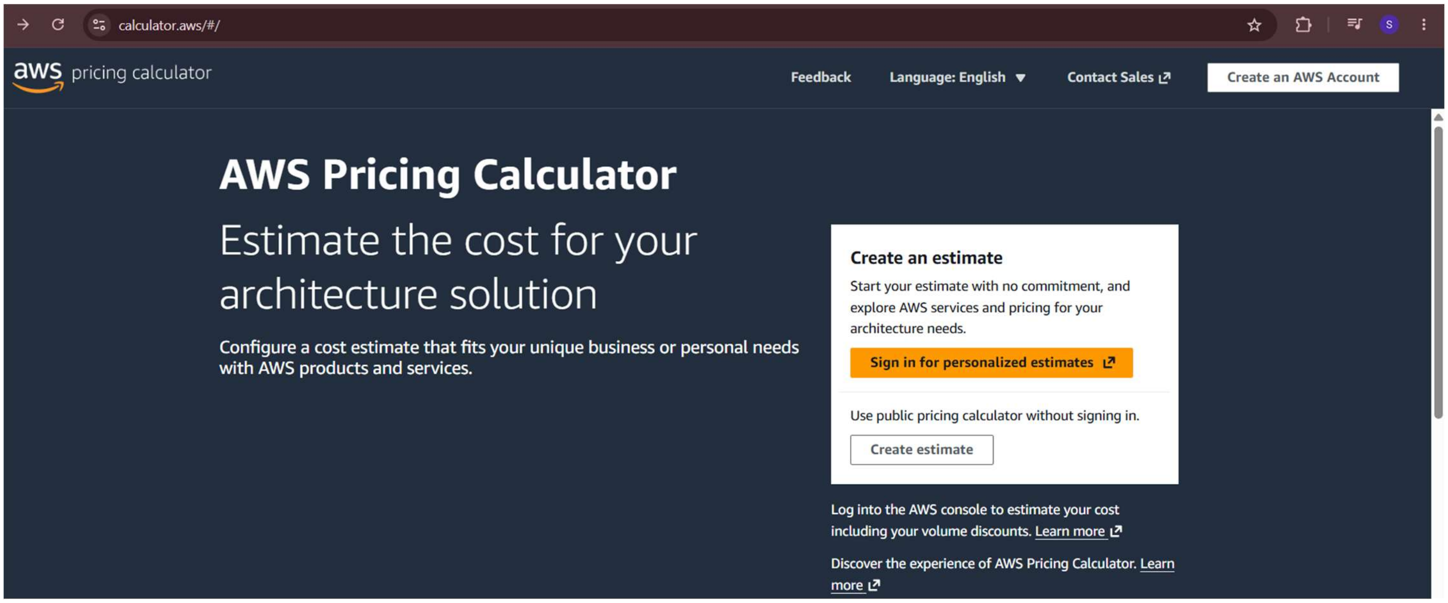
- Introduction to EC2
- Instance types explanation
- Scalability
- Instance Life Cycle
- Explore Aws MarketPlace



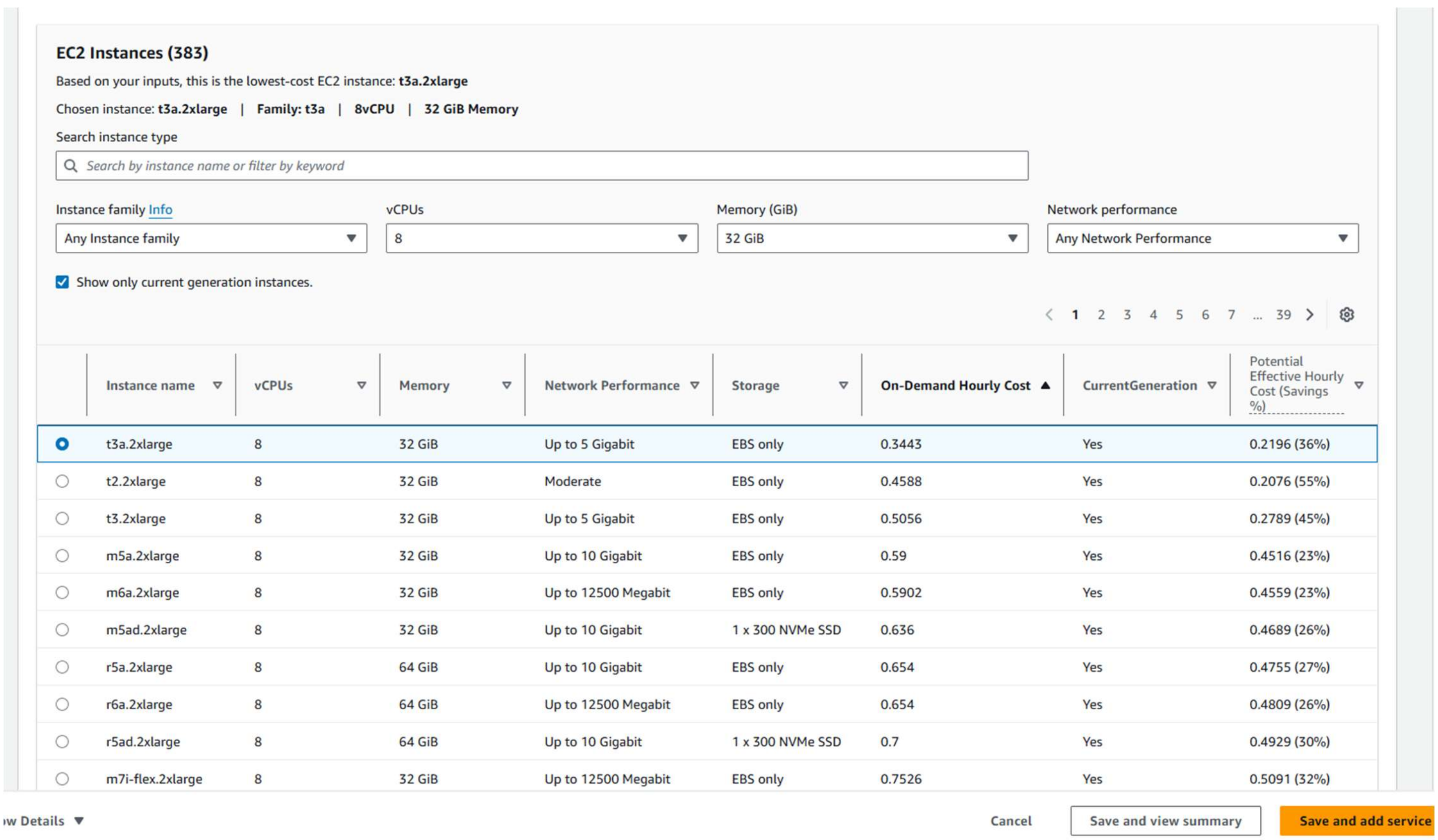
- Get instance screenshot (show present condition of instance)
- Key Pairs Explanation (key cant be changes after creating instance)
- AMI is stored in AWS and cost of storage is applied
- Intro to EBS
- Comparison of EBS vs Instance Store
    - ➢ Every server doesn't support Instance store

How to Calculate Costing

- Open https://calculator.aws
- Click create estimate



- Select region
- Find service and select its configuration and specifications and usage



- Calculated costs shown

Instance: 0.3443/Hour

Monthly: 251.34/Month

▼ Show calculations

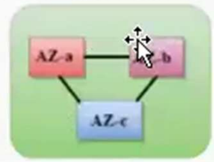
1 instances x 0.3443 USD On Demand hourly cost x 730 hours in a month = 251.339000 USD

On-Demand instances (monthly): 251.339000 USD

➢ Region and Availability Zones

Region

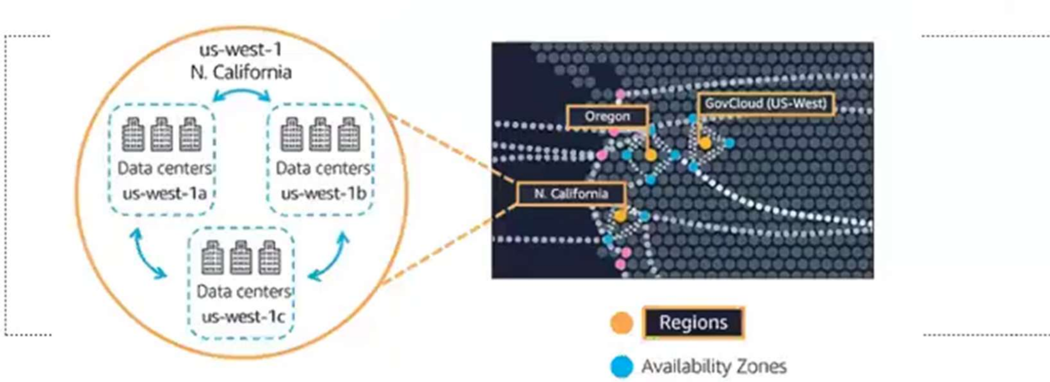
- A region is a geographical area. Each region consists of 2 more availability zones.
- A region is a collection of data centers which are completely isolated from other regions.
- A region consists of more than two availability zones connected to each other through links.



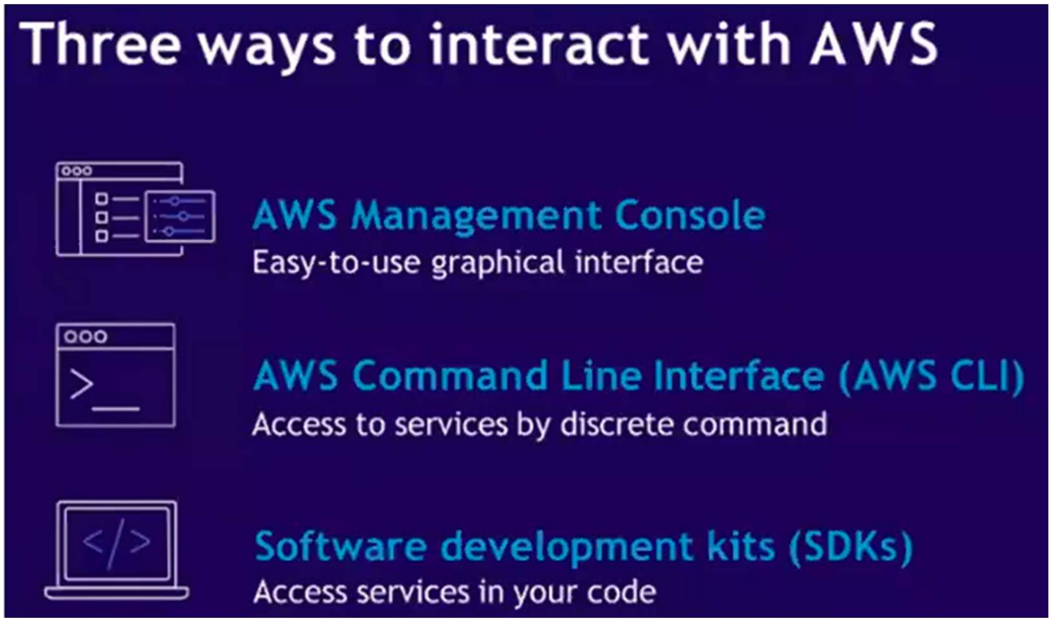
Availability zone as a Data Center

- An availability zone is a facility that can be somewhere in a country or in a city. Inside this facility, i.e., the Data Centre, we can have multiple servers, switches, load balancing, firewalls. The things that interact with the cloud sit inside the data centers.
- An availability zone can be several data centers, but if they are close together, they are counted as 1 availability zone.

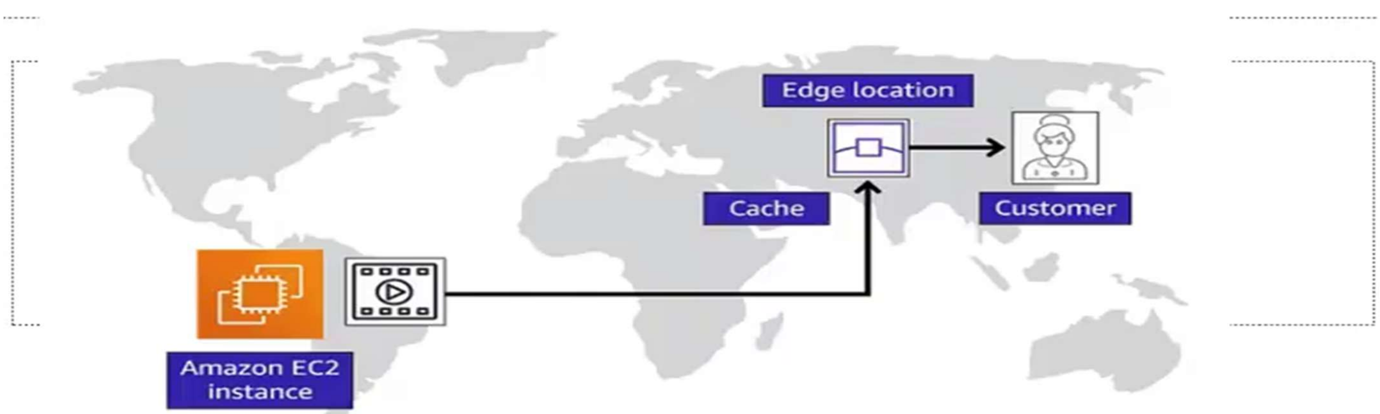
Understanding AWS Availability Zones



➢ Ways to connect with AWS



- **Edge Locations**



AMI(Amazon Machine Image)

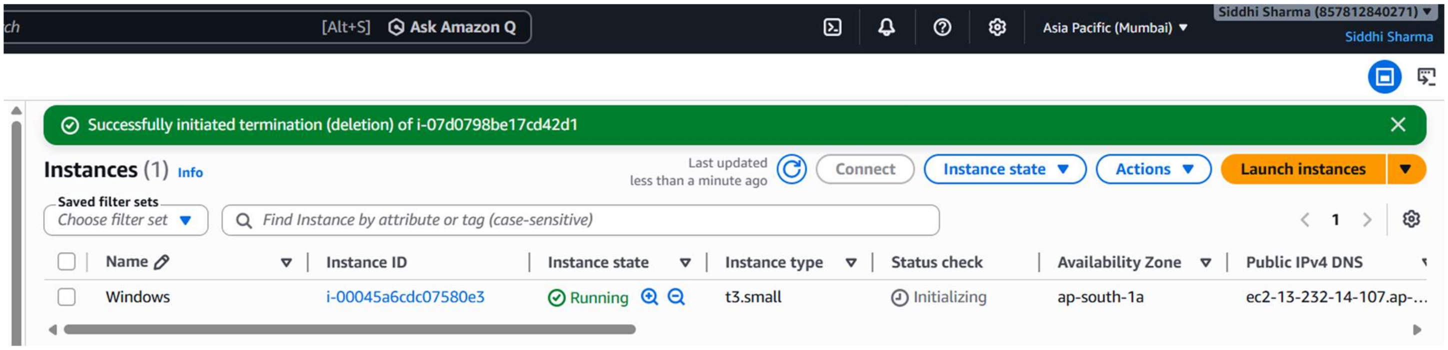
- An Amazon Machine Image (AMI) is a special type of virtual appliance that is used to create a virtual machine within the Amazon Elastic Compute Cloud (EC2).
- AMI is like an OS
- Like a Template (ready to create instance)
- An AMI is essentially a pre-configured template containing the necessary information to launch an instance in the AWS cloud. It can include the operating system, application server, applications, and related configurations. The introduction of AMIs significantly streamlines the process of server provisioning; allowing businesses to launch identical, consistent environments quickly and efficiently.

- **AMI Architecture and Fundamentals**
  - At its core, an AMI is a combination of several essential components. Understanding these components is key to mastering AMIs:
  - Operating System: The base software that runs on the instance, such as Amazon Linux, Ubuntu, or Windows.
  - Application Server and Software: Includes any necessary application software, databases, or frameworks.
  - Root Volume Snapshot: A point-in-time backup of the instance's root volume (i.e., the operating system and system configurations).
  - Launch Permissions: These define who can launch an instance from the AMI. Permissions can be set for individual users or groups within AWS

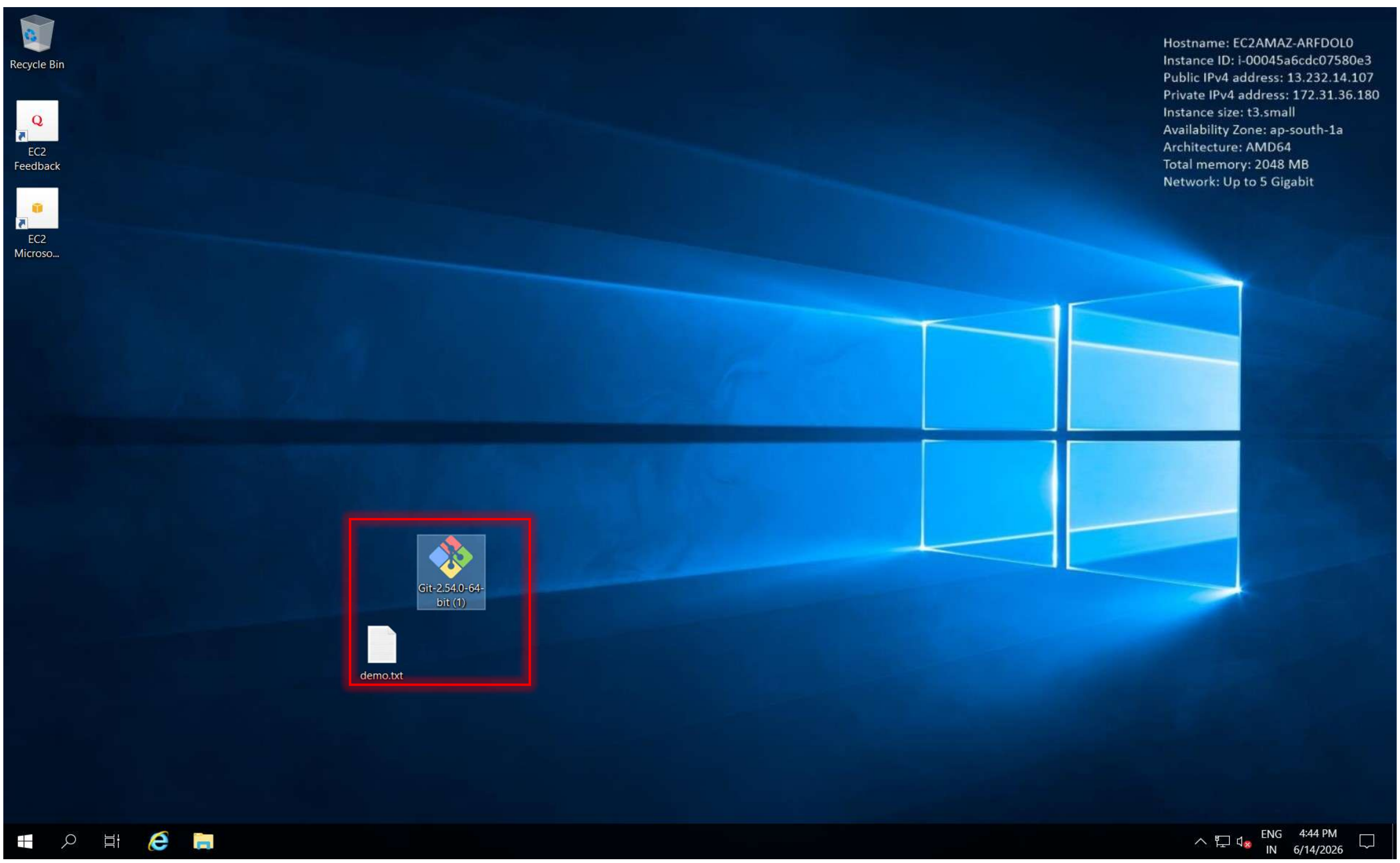
- **Types of Images**
  - Public: an AMI that can be used by anyone.
  - Paid: a for-pay AMI that is registered with Amazon MarketPlace and can be used by anyone who subscribes for it. MarketPlace allows developers to mark-up Amazon's usage fees and optionally add monthly subscription fees.
  - Shared: a private AMI that can only be used by Amazon EC2us ers who are allowed access to it by the developer

AMI(Amazon Machine Image) - LAB

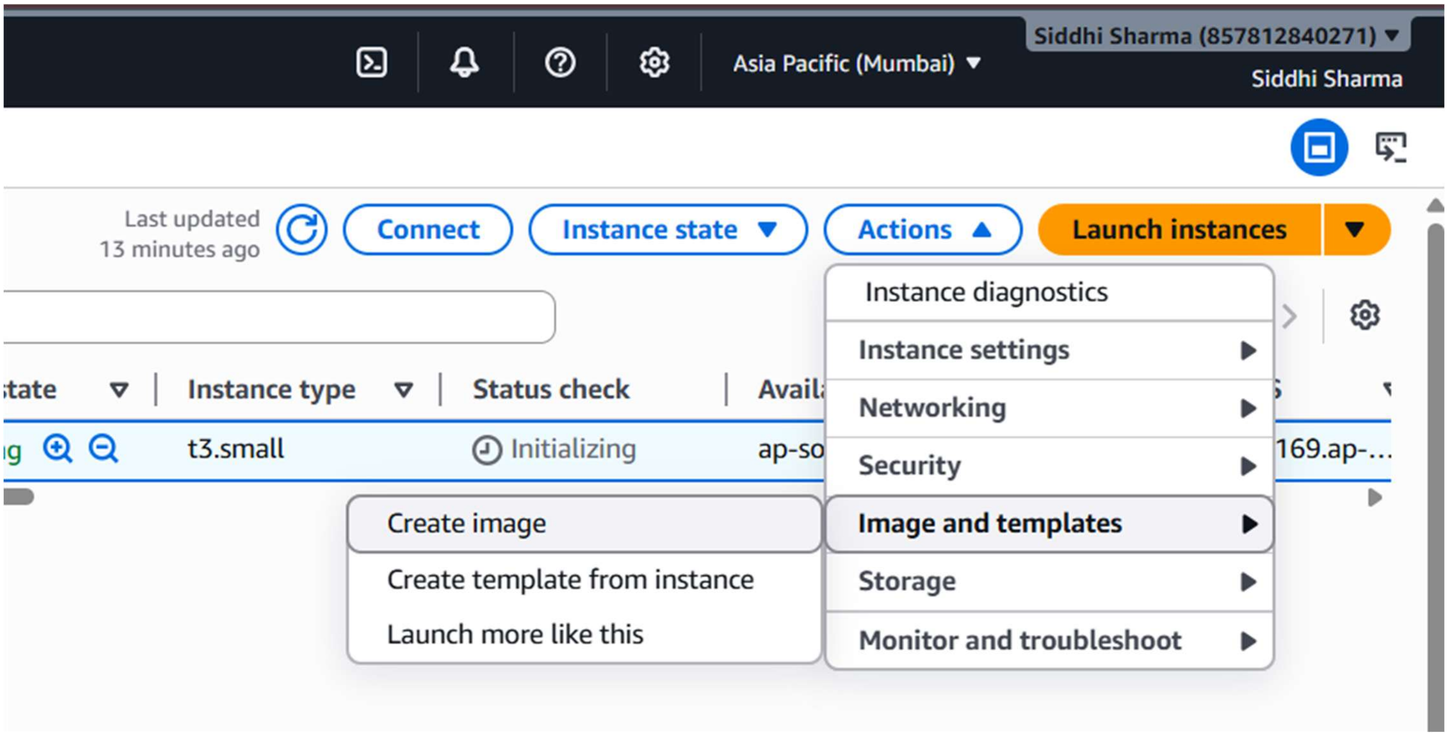
1. Create a new Windows Instance



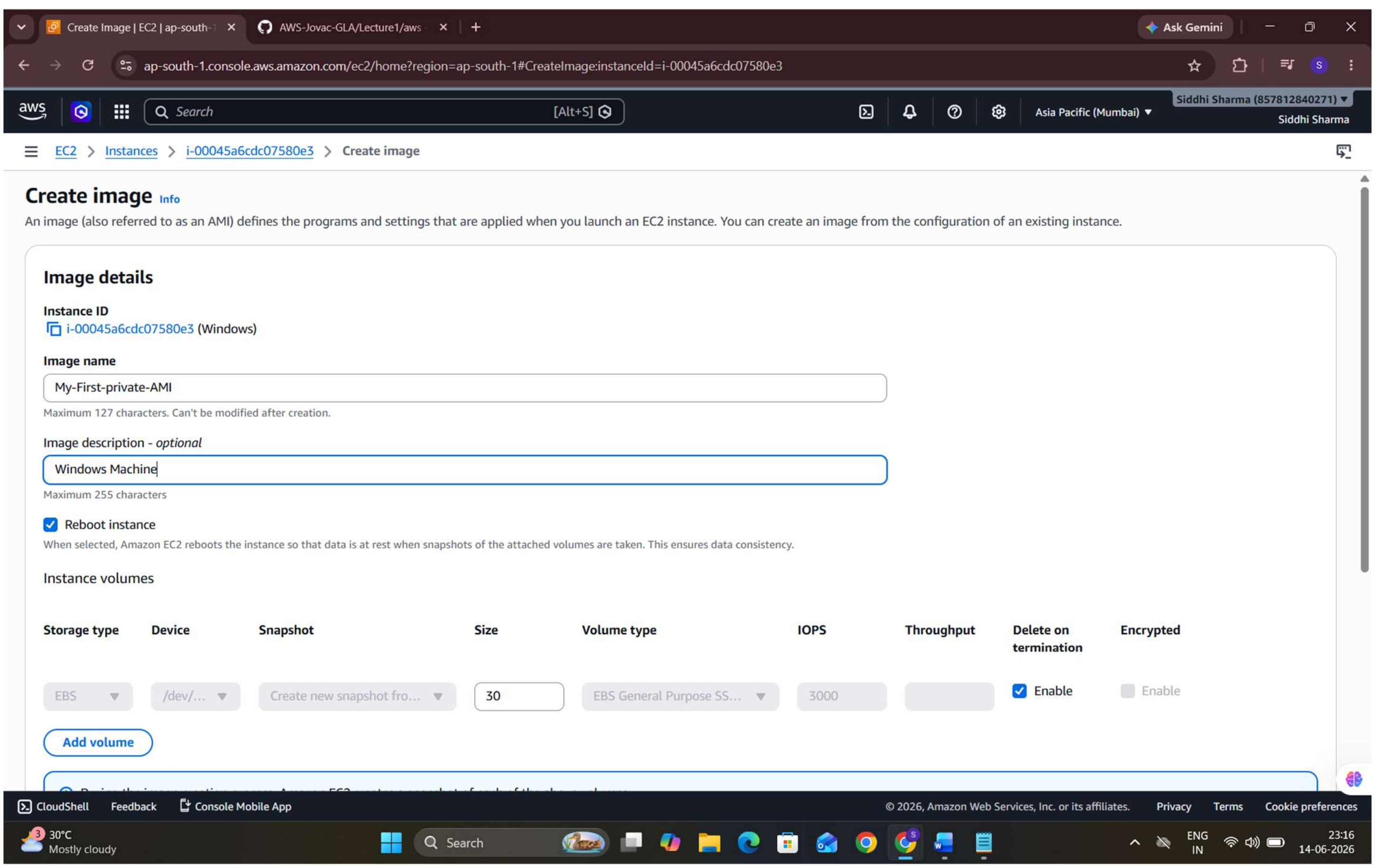
2. Access server using RDP client (Use steps in previous Lecture to get password and get access to server.
3. Enter ID Password
4. Create a TXT file in server
5. Install a Software(GITBASH) in server by copying setup to server



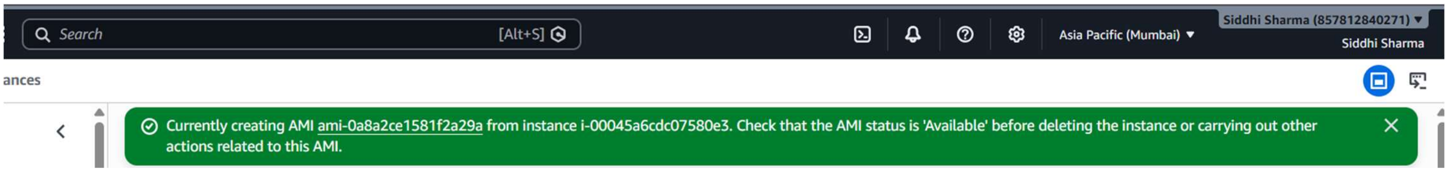
6. AWS suggest to stop the server whose you want to create AMI of.
7. Go to actions ->images and templates -> Create Image



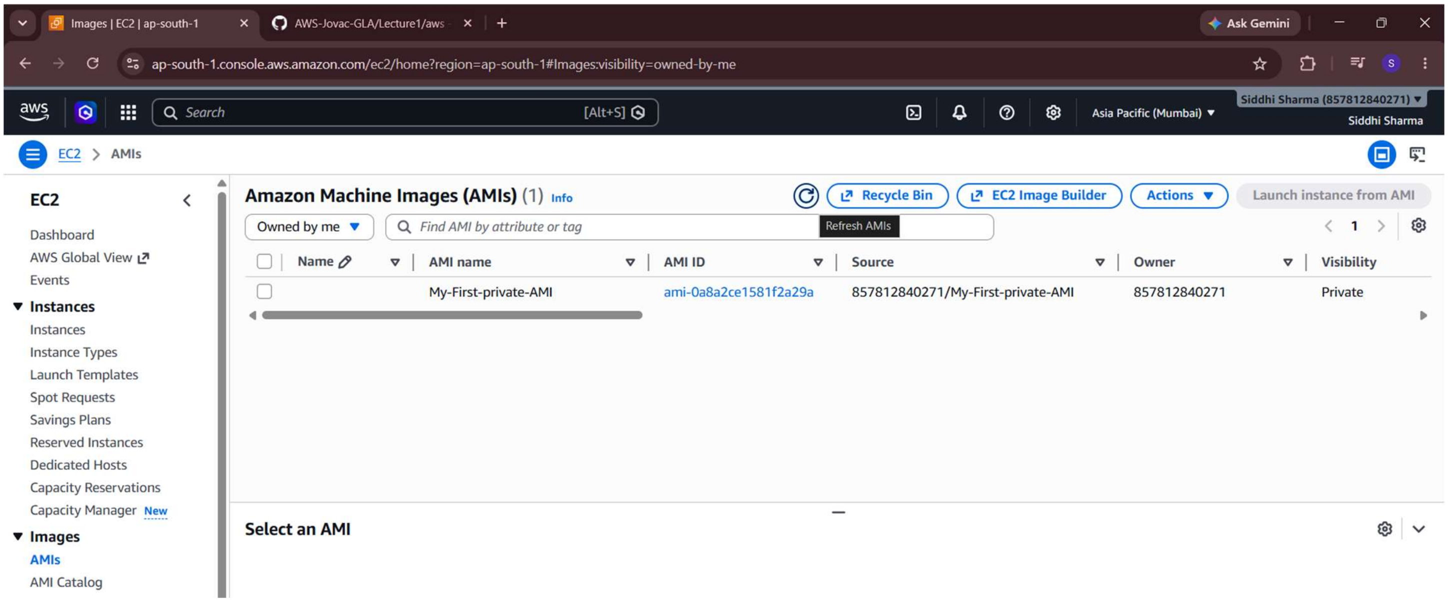
8. Create image window will open
9. Select AMI name



10. Check or uncheck Reboot option according to requirement
11. Create image par click kare to start creating



12. Ami created



13. Deregistration protection is about deleting AMI

- **User can enable Termination protection by going to Action and then Instance Settings to enable or disable it**

14. Go to EBS -> snapshots check percentage completed